

Experimental pipeline: confirmatory PCA vs. exploratory raw trajectories

Confirmatory (PCA)

Exploratory (Raw)

Prompt

LLM generation (Qwen2.5-7B-Instruct)

Token-level hidden-state trajectory $\mathbf{h}_t \in \mathbb{R}^{3584}$

Exploratory (ABL-1)

Raw 3584D hidden states
(no dimensionality reduction)

Confirmatory (pre-registered)

Variance-maximizing PCA

64D (PC1--64) / PC4--67
explained var. PC1--64 = 0.99992

κ -derived trajectory features
mean / max / p_{95} / std / AUC-density

Classifier → test AUROC

Comparison to response-length baseline (BL-LEN)